Corporation claimed to own 20,500 acres, containing 720,000,000 tons of coal on Graham Island, which it estimated at $360,000,000, although the whole property was to cost only about a three hundred and sixtieth part of that, and only three payments had been made on account of it. The next largest asset was stated to be 1500 acres at Quatsino, estimated to contain 20,000,000 tons of iron ore worth $2 per ton, or $40,000,000. The third asset in importance was stated to be 7,500,000 tons on Barclay Sound and Pitt Island, making another $15,000,000. This beats Lawson's Standard Oil exposures for high finance.

The most regrettable thing about the concern is that it was so intimately mixed up with British Columbia interests, and that leading men of Vancouver should have lent their names to bolstering up such wild finance. The inclusion of their names on the board was probably intended to give the concern an air of responsibility and respectability when it went to London for capital on such an infamous prospectus; but that does not free these gentlemen from the discredit of allowing their names to be used to mislead the public, and to bolster up a rotten concern. They probably received some stock consideration for the use of their names, but that is no excuse, and they cannot escape the odium attaching to having their names connected with the smash. No man should lend his name to a directorate unless he intends accepting the responsibility of the management of the company, and if he allows his name to remain on the board he must accept the full responsibility for the mismanagement and misrepresentation of the concern in connection with which his name so appears.

VANCOUVER ISLAND POWER COMPANY'S NEW POWER PLANT.

It has always been recognized that the City of Victoria has been badly located so far as a convenient water supply and source of hydro-electric power are concerned. The water supply has hitherto been the worst of any city on the Pacific Coast outside San Francisco. Although Goldstream has been available for 15,000,000 gals. daily, and as a limited source of water for power, the fact that it was owned by a private company with whom the City Council failed to make an agreement acceptable to the citizens has delayed the improvement of the civic water service until a supply can be obtained from Sooke Lake.

The question of power has been a more difficult matter to solve. Under the advice of the engineer, Wynn Meredith, who successfully carried through the Buntzen Lake installation of the B. C. Electric Railway Company at Vancouver, the Vancouver Island Power Company, a subsidiary organization of that company, placed engineers in the field for three years exploring all possible sources of power supply within fifty miles of Victoria. As a result it was decided to adopt the Jordan River as a source of power for operating the electric railway system of the city and suburbs, and for the supply of power and light for civic and industrial purposes, and for domestic use.

THE JORDAN RIVER.

This river takes its rise in the high mountains forming the backbone of Vancouver Island, on the watershed between the Cowichan Valley and the West Coast. The river enters the Straits of San Juan de Fuca at a point forty miles from Victoria, and from its mouth to its source at the Jordan Meadows, is about 24 miles in length. Its volume is increased by numerous tributaries, draining an area subject to heavy rainfall and deep snow, with long, wide valleys capable of being formed into immense reservoirs, from which can be furnished a supply of power sufficient to meet the needs of the city and suburbs for many years to come. The discovery of a source of power with such natural advantages for providing for the prospective demands of the future is a matter on which the Vancouver Island Power Company are to be congratulated. For a distance of six miles, from within a mile of its outlet to the heart of the mountains, the Jordan passes through a rugged, rocky defile, through which it must have appeared impossible to the uninitiated that a power line could be laid. But engineers scoff at such difficulties, and the visitor to the plant sees a well-built pipe line laid from the power station on the river flat near the sea shore to the lower reservoir, and a tramway and flume constructed along the rocky cliffs on the mountain side for a distance of six miles to the head of the present flume.

HOW THE WORK WAS CARRIED OUT.

A. B. Carey was the engineer in charge of the original exploration, and when the work was started E. Carpenter was placed in charge of it as resident engineer. Main camps for the accommodation of the workmen were established on the beach, at the mouth of the river (known as No. 1); at the lower dam, two miles from the power house (known as No. 2); at a point seven miles further up the river (known as No. 5), and at Bear Creek, 10 miles up the Jordan River, just beyond where it turns to the northwest (known as No. 10). A tramway was constructed from the site of the power house to Camp 5, where is the head of the present flume. The tramway between the power-house site and the top end of the pipe line, a distance of a mile and a half, is on a steep incline, and operated by winding engines and cables, one at the power house and the other at the top of the incline.

The remainder of the tramway follows the easy

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grade of the flume, so that the cars running over it can be operated by horses. The tramway is seven miles in length altogether. From the upper end of the tramway a wagon road was constructed to the site of the big dam on Bear Creek, four miles further on, the country here allowing of an almost level grade.

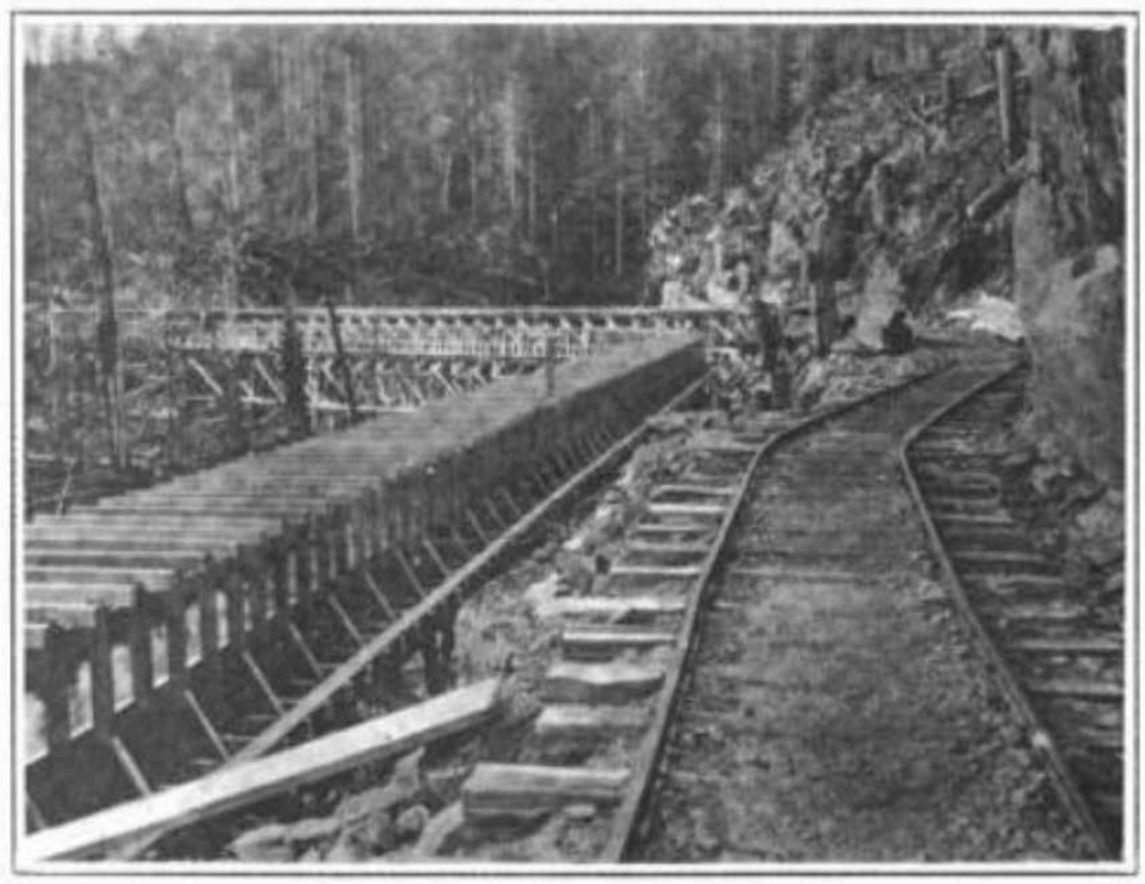
THE FLUME.

The flume is six feet wide by five feet in depth, with a grade of 6 per cent, and extends from the lower or forebay dam to Camp 5, where it heads in the upper dam on the Jordan River. This dam is 50 feet in length with overflow for flood water. The diversion of the water from Jordan River at this point is supplemented by another flume from Wye Creek, an important tributary coming in from the west. The main flume traverses the canyon for a length of six miles, supported on strong trestle works and following the sinuosities of the river, the path for the flume and tramway being at times hewn out of rocky promontories, or, supported by trestles, crossing gulches scores of feet in depth.

THE DAMS

Are four in number. The lower dam or forebay is situated at the top of the ridge overlooking the sea, and is at an elevation of 600 feet above the river at this point. This dam closes the mouth of a former river basin, and makes a reservoir capacity of 15 acres to be held in reserve in case of a break down of the flume. Two earthen dams of a height of 40 feet have been constructed here. The second dam is at the point of diversion of the water from the main river by means of the flume. The third dam is at the head of the Wye Creek supplementary flume, and the fourth dam is in course of construction on Bear River.

This latter dam is an immense undertaking, the dam being 997 feet in length, 50 feet in height, and 50 feet wide at base. It is intended for storage purposes, and is so located as to secure the water from a large catchment area, where snow lies late in the year—to July at least. The width of the valley and its great length of almost level stretch make it a remarkable natural reservoir site. To construct the dam a hydraulic plant has been installed, and the earth and gravel from which it will be constructed will be washed from a hill on the west side, and conveyed by a flume built over the site of the dam. Test pits have been sunk on the hill which is to be washed down for this purpose, and the adaptability of the ground to hydraulicing was proved before the installation of the plant. The base of the dam was sunk 20 feet below the bed of the creek and filled with clay. Its construction involves the removal and placing some 200,000 yards of earth. When completed this dam will give a storage capacity of about 400 acres, and by the construction of subsidiary dams along the valley of Bear River, this storage capacity can be increased to an immense extent. W. W. Alberts, of the Lake Buntzen staff, is in charge of the work.



Flume and Construction Tramway of Vancouver Island Power Company Ltd.

THE PIPE LINE.

The pipe line between the lower dam and the power house is two miles in length. It consists of a combination of lap-welded pipe of German manufacture, and double rivetted lap, supplied by the Jens-Orten-Boving Company, of London, Eng., through their agent for British Columbia, Mr. Kent, and their local engineer, D. L. A. Aplin. The size of

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the pipe is graduated from 44 inches to 30 inches in diameter, and by means of a Y, two pipe lines may be carried to the power house when required, though one has been laid for the present, and will suffice for some time. This pipeline delivers the water to the wheels of the power-house with a fall of 1150 feet, and a pressure of 500 lbs. to the square inch.

THE POWER HOUSE

is built of reinforced concrete, and its foundations have been sunk to a depth of 40 feet. Although these foundations were laid on what appeared to be a solid formation, and great precautions were taken to ensure stability, it was found after it had been erected that the north end was sinking. A drilling plant was secured, and holes drilled, with the result that the depression in the building was found to be due to a bed of peat, 12 feet thick, underlying the stratum on which the foundations were laid. This shows the old bed of the Jordan River to have been about 100 ft. lower than the present level, and can only be accounted for by the recent sinking of the west coast of Vancouver Island. Several Indian relics were found in the sinking of the foundations. Steps have been taken to strengthen the power-house by driving steel piling to bedrock and reinforcing it with concrete, the necessity for this work delaying the time by which it had been expected to supply power.

THE POWER PLANT

consists of two generating units, direct connected to Pelton wheels, transformers, etc. The unit in readiness for operation has a capacity of 6,000 h.p. From this the current will be stepped up to 35,000 volts by means of transformers. For conveying it to the city of Victoria and Saanich Peninsula, a line of 40 miles in length is required, and this has been constructed of strong poles, carrying three cables of aluminium wire, well insulated, and supported by metal cross-arms. The transmission line follows for the most part the Sooke road, a branch line running off from Victoria to Saanich and the Vancouver Portland Cement Works at Tod's Inlet. These works alone have contracted for 1,500 h.p. as soon as it is available. The transmission line has been constructed through one of the heaviest belts of timber in British Columbia, the work of felling the timber and clearing the right-of-way having been one of the heaviest parts of the undertaking. The electrical plant is supplied by the Allis-Chalmers Bullock Company.

To provide the lumber for the tramway, flume and buildings, a sawmill of a capacity of 30,000 feet a day was installed at Camp 2, and was run for a year at full capacity.

The Jordan River plant provides in its initial development, for 6,000 h.p., and will later be increased to a maximum of 24,000 h.p. This will prove an important factor in the extension of electric railway construction in Victoria and suburbs, and also in the development of industries, which will be promoted by a reliable supply of cheap power.

The plant went into operation on Sept. 16th.

THE LONG TUNNEL AT GREENWOOD.

The Greenwood Phoenix Tramway Company, which undertook the boring of the long tunnel between Greenwood and Phoenix via the Boundary Creek Valley, has made good progress, going ahead for a time at the rate of ten feet a day, and at last accounts was in over 3,000 feet. The stockholders, who are mostly Chicago men, advised the manager, Duncan McIntosh, that they would provide the funds to drive a mile at present. Twenty-two men and two 3¼ inch drills have been employed on the work. The distance between the two points is 17,000 feet, and the vertical depth under Phoenix is 2000 feet. The idea is to cut mineral veins outcropping on claims between.

In May, 1909, the city of Greenwood passed a by-law bonusing the scheme to the extent of $50,000, and the work was started in July of last year. The size of the bore is 9 feet by 8 feet, and it is equipped with a double track.

In its progress the tunnel cut several small veins, some having values and some having none. The principal strike was at 1,800 feet from the portal, and at a depth of 818 feet under the Defiance claim, where a vein was crossent showing values of $110 per ton. The Strathmore, Providence and Elkhorn ground has yet to be crossed.

The company has consolidated between 20 and 30 claims, aggregating 800 acres, along the course of the tunnel, and thus owns 6,000 feet of the ground traversed. If the tunnel develops sufficient ore, the company proposes to instal a smelter.

A station has been cut and preparations made to develop the Defiance vein already crossent. The ore assays .31 gold, 191.30 silver, and 2.7 copper, indicating that some of these ore bodies at least hold their values with depth.

MINE RESCUE WORK.

With the rapid development of mine rescue work in the last two years, and the increasing use of the mine-breathing apparatus, there has come to the Federal Bureau of Mines at Washington a demand for information concerning this apparatus, and how it should be properly used. In response the Bureau of Mines has issued Miners' Circular No. 4, on "The Use and Care of Mine-rescue Breathing Apparatus," by James W. Paul, who has general charge of the rescue work.

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